

Khaled Lakhdher

Computer Engineer | AI & Data Science | Full-Stack AI Developer

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Portfolio: khaled-lakhdher-portfolio.vercel.app

PROFILE

Passionate Machine Learning engineer with a solid foundation in deep learning, computer vision, and natural language processing. Proficient in Python and TensorFlow, with hands-on experience building and deploying AI models. Eager to apply my skills to real-world challenges and contribute to innovative artificial intelligence solutions.

EXPERIENCE

AI Engineer - Final-Year Project (PFE)

Feb 2026 - June 2026 · Tunisia

CTT - Carthage Training & Technologies

- Designed and developed **Hotel Mind**, a conversational hotel-booking assistant on a grounded RAG architecture connected to a real-time SOAP API, with a hallucination-mitigation layer that verifies every factual claim.
- Delivered a containerized full-stack solution reaching nDCG@5 0.82, MRR@5 0.90, and a 100% success rate on a benchmark query set.

Artificial Intelligence Developer Intern

July 2025 - September 2025 · Hammamet

CTT - Carthage Training & Technologies

- Fine-tuned a Hugging Face NLP model to classify customer reviews by sentiment (positive, negative, neutral).
- Built a Django backend to manage customer data and automate export of results to Excel for reporting.
- Handled all preprocessing, training, and integration scripts in Python.

PROJECTS

Nexus AI - AI-Employee Management Platform

- Built a multi-tenant platform (FastAPI, PostgreSQL/pgvector, Redis, Next.js) governing LLM agents with IAM permissions, vector memory, multi-agent orchestration over a Redis message bus, and real-time WebSocket monitoring, behind a provider-agnostic model gateway (Anthropic, OpenRouter, Gemini).

Personalized Voice Assistant

- Fine-tuned Whisper (STT) and SpeechT5 (TTS) with CSM-1B voice cloning to build a personalized voice assistant, cutting response time by 30%.

CodeReview-Eval - Benchmarked LLM Code Reviewer

- Built an LLM code-review and evaluation framework on the BugsInPy benchmark (500 real bugs); using an LLM-as-judge, showed a single detection score overstates capability by $\sim 1.8\times$ (78% localization vs. 43% diagnosis) across 3 model tiers, with a live Streamlit demo.

CERTIFICATIONS

- Fundamentals of Deep Learning - NVIDIA (Nov 22, 2022)
- Building Transformer-Based Natural Language Processing Applications - NVIDIA (Dec 19, 2025)
- Applications of AI for Predictive Maintenance - NVIDIA (Oct 27, 2025)

SKILLS

Programming Languages: Python · Java · C · SQL · JavaScript · TypeScript · React.js · React Native · HTML/CSS · Tailwind CSS

AI Libraries & Frameworks: Transformers · PyTorch · TensorFlow · Keras · Hugging Face · Hugging Face Hub · scikit-learn · Pandas · NumPy · Matplotlib

Development Frameworks & Tools: Gradio · Streamlit · Flask · Django · Zeep · LangSmith · Pydantic · Docker Compose · Nginx / Reverse Proxy · Postman · Git · GitHub

Concepts & Technologies: Artificial Intelligence (AI) · Machine Learning (ML) · Deep Learning (DL) · Large Language Models (LLMs) · AI Agents · Agentic AI · Tool-Calling · Prompt Engineering · LLM Evaluation · Benchmarking · LLM-as-a-Judge · NLP · ASR · GANs · Vector Databases · Semantic Search · Embeddings · Information Retrieval · API Development · NoSQL · Cloud Computing

EDUCATION

Computer Engineering - AI & Data Science Specialization

2023 - 2026

EPI - International Polytechnic School · Sousse, Tunisia

Preparatory Cycle

2021 - 2023

EPI - Private International Polytechnic School · Sousse, Tunisia

LANGUAGES

French: DELF B2

English: TOEIC (Listening & Reading) - 825/990